

PEER REVIEW REPORT

Manuscript Title: Atomic-Layer-Deposited ZnF₂ on Aluminum Enabling in situ NaZn₁₃/NaF Interphase for Robust Anode-Free Sodium Batteries

Authors: V.P. Nguyen, A. Sarapulova, M. Guenthel, M. Knaebbeler-Buss, C. Kim, Y.-W. Byeon, M. Lee, K. Yim, E. Troester, S. Dsoke, S.-M. Lee

Version / Date: Ver_6_20260319

Review Date: 2026-03-20

1. Manuscript Overview

Field	Details
Keywords	anode-free sodium battery, sodiophilic, NaF-rich SEI, cycling stability, dendrite
Core Claim	ALD-derived ZnF ₂ on Al converts in situ to NaZn ₁₃ /NaF interphase → ~99.7% CE, >1000 h symmetric cell stability, 323 Wh kg ⁻¹ in anode-free Na NVP cells
Affiliations	KIMM (Korea), Fraunhofer ISE (Germany), FMF Freiburg (Germany), KIST (Korea), KIER (Korea), CNU (Korea), ALU-Freiburg (Germany), UST (Korea)

2. Fatal Error Pre-Screening

WARNING: Four fatal/critical issues were detected during pre-screening.

#	Category	Fatal Error Description	Location
F-1	Data Inconsistency	NVP cathode mass loading: 11.5 mg/cm ² (Abstract) vs. 13.2 mg/cm ² (Para [70], lean-Na) vs. 11.6-12.2 mg/cm ² (Para [72], AFSB). Three different values with no explanation.	Abstract vs. Para [70] vs. Para [72]
F-2	Critical Typo - Microscopy Term	"annual dark field scanning TEM (ADF-STEM)" must be "annular dark field." ADF = Annular Dark Field, not Annual Dark Field. Fundamental microscopy terminology error.	Para [54]; Figure 3 caption
F-3	Sentence Fragment - Grammar	"Although modifying Al...can lower the nucleation barrier.[10-13] the resulting SEI often remains fragile..." - the Although subordinate clause is ended with a period, creating a sentence fragment without a main	Para [33], Introduction

		clause.	
F-4	Spelling Error - Proper Name	"Emmy-Noether-Strasse" should be "Emmy-Noether-Strasse" (named after mathematician Emmy Noether).	Affiliation [g]

3. Detailed Peer Review

3.1 Originality | Rating: Moderate-High (4/5)

Strengths:

- First ALD-ZnF₂/NaZn₁₃/NaF interphase for AFSBs; well-differentiated from prior slurry-cast approaches.
- Clear novelty statement relative to Wang et al. (Ref. 23) slurry-cast MF_x@Al.
- "First example" claim appears well-supported by the literature survey.

Areas Requiring Improvement:

- The Introduction does not discuss precedents from ALD coatings in Li-metal batteries. A brief comparison would contextualize why ALD has not been previously applied in Na systems.
- The N/P ratio of ~0.33 in the lean-Na full-cell (Para [70]) is extremely low. The practical significance should be discussed explicitly.

3.2 Creativity | Rating: Moderate-High (4/5)

Strengths:

- Dual-function ZnF₂ design is logically motivated: Zn -> sodiophilic NaZn₁₃; F -> NaF-rich SEI.
- DFT adsorption energy calculations on NaZn₁₃ (100) and ZnF₂ (110) provide quantitative design guidance.
- Cryo-STEM/EELS cross-sectional analysis (Figures 3D, 3E) provides direct structural evidence for conversion.

Areas Requiring Improvement:

- The layered structure (NaZn₁₃ at electrolyte side, unreacted ZnF₂ near Al) is intriguing but not fully explained mechanistically. Why does NaZn₁₃ segregate to the outer layer?
- The long-term stability of the NaZn₁₃ phase during repeated cycling is not discussed.

3.3 Data Objectivity | Rating: Moderate (3/5)

Strengths:

- Multi-technique evidence (cryo-STEM, EELS, XPS depth profiling, DFT) converges on a consistent interphase model.
- Symmetric cell stability of >1000 h at 1 mA cm⁻² is a strong, concrete metric.

Areas Requiring Improvement:

- CE of ~99.7% reported without $n \geq 3$ cell statistics or error bars. Mean +/- SD must be provided.
- Energy density of 323 Wh kg⁻¹: calculation basis is ambiguous for an anode-free cell (cathode mass only?).
- Na 1s XPS peak assigned to "metallic Na⁰ in NaZn₁₃" lacks reference binding energy from literature.
- ADF-STEM contrast disclaimer (Para [54]) must also appear in Figure 3D caption.

3.4 English Language Quality

#	Location	Original Text + Issue	Suggested Correction
E-1	Abstract, Para [28]	"...~99.7% CE by a depositing/stripping capacity..." Wrong preposition "by a"	"...at a depositing/stripping capacity of 1 mAh cm ⁻² "
E-2	Para [33], Intro	"...nucleation barrier.[10-13] the resulting SEI..." Period creates sentence fragment after "Although" clause	Change period to comma: "...nucleation barrier,[10-13] the resulting SEI often remains fragile."
E-3	Para [35], Intro	"BiF ₃ -derived interphases - forming Na ₃ Bi and NaF - as effective" Hyphens used instead of em-dashes	"BiF ₃ -derived interphases - forming Na ₃ Bi and NaF in situ -- as particularly effective"
E-4	Para [54], Results	"cross-sectional ANNUAL dark field...ADF-STEM" CRITICAL TYPO: annual -> annular	"cross-sectional ANNULAR dark field scanning TEM (ADF-STEM)"
E-5	Para [61], Results	"amplified by the tip effect" Unnecessary scare quotes around standard technical term	Remove quotes: amplified by the tip effect
E-6	Para [67], Results	"both anode-less and anode-free sodium battery systems" Two similar undefined terms used without distinction	Define at first use: "...both lean-Na (anode-lean) and truly anode-free configurations"
E-7	Affiliation [g]	"Emmy-Noether-Strasse" Misspelling of Emmy Noether surname	"Emmy-Noether-Strasse"
E-8	Para [62], Results	"residual ZnF ₂ likely acted as sodiophilic nucleation sites" Role of residual ZnF ₂ should be explicitly justified	Refer to Figure 1C DFT data or clarify ZnF ₂ role vs NaZn ₁₃ role.
E-9	Abstract, Para [28]	"NaZn ₁₃ is expected to enhance...NaF strengthens...ensuring Na ⁺ flux" Inconsistent modal hedging	Use uniform declarative language: if data supports claims, use consistent declarative tense throughout.
E-10	Para [50], Results	"cycled five times between 0.01 and 1.0 V to stabilize SEI" Unusually wide upper cutoff; rationale not stated	Add brief justification for 1.0 V upper cutoff in Experimental section.

3.5 Figure and Data Issues

#	Figure / Location	Severity	Issue Description
---	-------------------	----------	-------------------

FIG-1	Abstract / Para [70] / Para [72]	CRITICAL	NVP mass loading discrepancy: 11.5 (Abstract), 13.2 (lean-Na), 11.6-12.2 mg/cm ² (AFSB). Must be reconciled.
FIG-2	Figure 3D / Para [54]	CRITICAL	"annual dark field" -> "annular dark field." ADF = Annular Dark Field. Fundamental error.
FIG-3	Figure 3D caption	MAJOR	ADF-STEM contrast disclaimer (diffraction + mass-thickness, NOT Z-contrast) appears only in main text, not in figure caption.
FIG-4	Figure 5A / Para [65]	MAJOR	CE ~99.7% reported without n>=3 cell statistics or error bars. Mean +/- SD from multiple cells required.
FIG-5	Figure 6E / Para [72]	MAJOR	Energy density 323 Wh/kg: calculation basis ambiguous for anode-free cell. Provide explicit formula.
FIG-6	Figure 3H, 3I / Para [57]	MODERATE	Na 1s XPS assigned to "metallic Na ⁰ in NaZn ₁₃ " without literature reference binding energy value.
FIG-7	Figure 4A / Para [60]	MODERATE	"shows smooth stripping profiles" - present tense used inconsistently within past-tense Results section.
FIG-8	Figure S9 / Para [61-62]	MINOR	Na layer thickness: 26 um (bare Al) vs 23.6 um (ZnF ₂ /Al). ~10% difference may not be significant. Report mean +/- SD.

3.6 Figures Requiring Priority Revision

Figure 3D -- ADF-STEM Cross-Section

Current State: Image labeled "annual [sic] dark field." Mixed diffraction+mass-thickness contrast caveat missing from caption.

Required Actions: 1. Correct "annual" to "annular" in all occurrences.
2. Add contrast disclaimer to Figure 3D caption.
3. Consider acquiring conventional HAADF-STEM for Z-contrast comparison.

Figure 6 -- Full-Cell Performance & Energy Density

Current State: Three different NVP mass loadings appear. Energy density formula for 323 Wh/kg not explicitly defined.

Required Actions: 1. Reconcile all mass loading values (or explicitly state different cathode batches).
2. Provide explicit energy density formula in caption.
3. Add footnote to Figure 6E specifying calculation basis for each compared work.

Figure 5A -- CE Statistics

Current State: Headline CE of ~99.7% from a single representative cell with no statistical validation.

- Required Actions:**
1. Measure CE on $n \geq 3$ independent cells.
 2. Report mean CE $\pm$ SD in figure, text, and abstract.
 3. If $n=1$, acknowledge this limitation.

4. Overall Assessment

Criterion	Rating	Key Comment
Originality	Moderate-High (4/5)	First ALD-ZnF ₂ /NaZn ₁₃ /NaF interphase for AFSBs; clearly differentiated from slurry-cast precedents
Creativity	Moderate-High (4/5)	Clever dual-function ZnF ₂ design; strong cryo-STEM/EELS mechanistic evidence
Data Objectivity	Moderate (3/5)	Mass loading inconsistency (critical); CE statistics incomplete; energy density basis ambiguous
English Quality	Moderate (3/5)	Critical typo (annual/annular); sentence fragment; several grammatical/stylistic issues
Figure Quality	Moderate (3/5)	Missing caption caveat for ADF-STEM; mass loading/energy density inconsistencies in Fig. 6
Scientific Rigor	Moderate-High (4/5)	Generally well-supported; Na 1s XPS assignment needs additional justification

5. Recommendation

MAJOR REVISION REQUIRED

Mandatory Revision Checklist

- [CRITICAL] Reconcile NVP mass loading: unify the three values (11.5 / 13.2 / 11.6-12.2 mg/cm²) across Abstract, lean-Na section, and AFSB section.
- [CRITICAL] Correct "annual dark field" -> "annular dark field" in Para [54] and Figure 3D caption.
- [CRITICAL] Fix sentence fragment in Para [33]: change period to comma -> "...nucleation barrier,[10-13] the resulting SEI often remains fragile..."
- [MAJOR] Report CE from $n \geq 3$ independent cells with mean $\pm$ SD; add error bars to Figure 5A.
- [MAJOR] Add ADF-STEM contrast disclaimer to Figure 3D caption.
- [MAJOR] Clarify energy density calculation basis for 323 Wh/kg with explicit formula; ensure consistent comparison in Figure 6E.
- [MODERATE] Provide Na 1s XPS reference binding energy for Na in NaZn₁₃ to justify metallic Na⁰ assignment.
- [MODERATE] Correct affiliation spelling: "Emmy-Noether-Strasse" -> "Emmy-Noether-Strasse".
- [MINOR] Replace hyphens with em-dashes in Para [35].
- [MINOR] Change "by a depositing/stripping capacity" -> "at a depositing/stripping capacity" in Abstract.
- [MINOR] Remove scare quotes around "the tip effect" in Para [61].

- [MINOR] Define "anode-less" vs. "anode-free" at first use in Para [67], or unify terminology.

6. Closing Note

This manuscript reports scientifically interesting and timely work on ALD-engineered interphases for anode-free sodium batteries. The core concept of using ALD-ZnF₂ to generate a dual NaZn₁₃/NaF interphase is creative and well-motivated, and the multi-technique characterization (cryo-STEM/EELS, XPS depth profiling, DFT) provides compelling mechanistic evidence. However, several critical issues must be resolved: the NVP mass loading inconsistency undermines reproducibility, the "annual/annular" typo damages scientific credibility, and CE statistics are insufficient for a headline claim. Addressing the mandatory revisions will substantially strengthen the manuscript and bring it to publication readiness.